

B2 Manipulate surds

Name: _____

Class: _____

Date: _____

Time: **242 minutes**

Marks: **203 marks**

Comments:

EXAM PAPERS PRACTICE

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Q1.

It is given that $\cos 15^\circ = \frac{1}{2}\sqrt{2+\sqrt{3}}$ and $\sin 15^\circ = \frac{1}{2}\sqrt{2-\sqrt{3}}$

Show that $\tan^2 15^\circ$ can be written in the form $a + b\sqrt{3}$, where a and b are integers.

Fully justify your answer.

(Total 3 marks)

Q2.

- (a) Determine a sequence of transformations which maps the graph of $y = \sin x$ onto the graph of $y = \sqrt{3} \sin x - 3 \cos x + 4$

Fully justify your answer.

(7)

- (b) (i) Show that the least value of $\frac{1}{\sqrt{3} \sin x - 3 \cos x + 4}$ is $\frac{2-\sqrt{3}}{2}$

(2)

- (ii) Find the greatest value of $\frac{1}{\sqrt{3} \sin x - 3 \cos x + 4}$

(1)

(Total 10 marks)

Q3.

Show that $\frac{5\sqrt{2}+2}{3\sqrt{2}+4}$ can be expressed in the form $m + n\sqrt{2}$, where m and n are integers.

(Total 3 marks)

Q4.

A circle with centre C has equation $x^2 + y^2 + 8x - 12y = 12$

- (a) Find the coordinates of C and the radius of the circle.

(3)

- (b) The points P and Q lie on the circle.
The origin is the midpoint of the chord PQ .

Show that PQ has length $n\sqrt{3}$, where n is an integer.

(5)

(Total 8 marks)

Q5.

A curve, C , has equation $y = x^2 - 4x + k$, where k is a constant.

It crosses the x -axis at the points $(2 + \sqrt{5}, 0)$ and $(2 - \sqrt{5}, 0)$

- (a) Find the value of k .

(2)

- (b) Sketch the curve C , labelling any intersections with the axes.

(3)

(Total 5 marks)

Q6.

- (a) (i) Express $\sqrt{18}$ in the form $k\sqrt{2}$, where k is an integer.

(1)

- (ii) Simplify $\frac{\sqrt{8}}{\sqrt{18} + \sqrt{32}}$.

(3)

- (b) Express $\frac{7\sqrt{2} - \sqrt{3}}{2\sqrt{2} - \sqrt{3}}$ in the form $m + \sqrt{n}$, where m and n are integers.

(4)

(Total 8 marks)

Q7.

A circle with centre $C(-3, 2)$ has equation

$$x^2 + y^2 + 6x - 4y = 12$$

- (a) Find the y -coordinates of the points where the circle crosses the y -axis.

(3)

- (b) Find the radius of the circle.

(3)

- (c) The point $P(2, 5)$ lies outside the circle.

- (i) Find the length of CP , giving your answer in the form $\sqrt{n}$, where n is an integer.

(2)

- (ii) The point Q lies on the circle so that PQ is a tangent to the circle. Find the length of PQ .

(2)

(Total 10 marks)

Q8.

- (a) (i) Express $\sqrt{48}$ in the form $n\sqrt{3}$, where n is an integer.

(1)

- (ii) Solve the equation

$$x\sqrt{12} = 7\sqrt{3} - \sqrt{48}$$

giving your answer in its simplest form.

(3)

- (b) Express $\frac{11\sqrt{3} + 2\sqrt{5}}{2\sqrt{3} + \sqrt{5}}$ in the form $m - \sqrt{15}$, where m is an integer.

(4)

(Total 8 marks)

Q9.

- (a) (i) Express $2x^2 + 6x + 5$ in the form $2(x + p)^2 + q$, where p and q are rational numbers.

(2)

- (ii) Hence write down the minimum value of $2x^2 + 6x + 5$.

(1)

- (b) The point A has coordinates $(-3, 5)$ and the point B has coordinates $(x, 3x + 9)$.

- (i) Show that $AB^2 = 5(2x^2 + 6x + 5)$.

(3)

- (ii) Use your result from part (a)(ii) to find the minimum value of the length AB as x varies, giving your answer in the form $\frac{1}{2}\sqrt{n}$, where n is an integer.

(2)

(Total 8 marks)

Q10.

- (a) (i) Simplify $(3\sqrt{2})^2$.

(1)

- (ii) Show that $(3\sqrt{2} - 1)^2 + (3 + \sqrt{2})^2$ is an integer and find its value.

(4)

- (b) Express $\frac{4\sqrt{5} - 7\sqrt{2}}{2\sqrt{5} + \sqrt{2}}$ in the form $m - \sqrt{n}$, where m and n are integers.

(4)

(Total 9 marks)

Q11.

Angle α is acute and $\cos \alpha = \frac{3}{5}$. Angle β is **obtuse** and $\sin \beta = \frac{1}{2}$.

- (a) (i) Find the value of $\tan \alpha$ as a fraction.

(1)

- (ii) Find the value of $\tan \beta$ in surd form.

(2)

- (b) Hence show that $\tan(\alpha + \beta) = \frac{m\sqrt{3} - n}{n\sqrt{3} + m}$, where m and n are integers.

(3)

(Total 6 marks)

Q12.

Express $\frac{5\sqrt{3} - 6}{2\sqrt{3} + 3}$ in the form $m + n\sqrt{3}$, where m and n are integers.

(Total 4 marks)

Q13.

- (a) Simplify $(3\sqrt{3})^2$.

(1)

- (b) Express $\frac{4\sqrt{3} + 3\sqrt{7}}{3\sqrt{3} + \sqrt{7}}$ in the form $\frac{m + \sqrt{21}}{n}$, where m and n are integers.

(4)

(Total 5 marks)

Q14.

- (a) (i) Given that $\tan 2x + \tan x = 0$, show that $\tan x = 0$ or $\tan^2 x = 3$.

(3)

- (ii) Hence find all solutions of $\tan 2x + \tan x = 0$ in the interval $0^\circ < x < 180^\circ$.

(1)

- (b) (i) Given that $\cos x \neq 0$, show that the equation

$$\sin 2x = \cos x \cos 2x$$

can be written in the form

$$2 \sin^2 x + 2 \sin x - 1 = 0$$

(3)

- (ii) Show that all solutions of the equation $2 \sin^2 x + 2 \sin x - 1 = 0$ are given by

$$\sin x = \frac{\sqrt{3} - 1}{p}, \text{ where } p \text{ is an integer.}$$

(3)

(Total 10 marks)

Q15.

- (a) (i) Express $\sqrt{48}$ in the form $k\sqrt{3}$, where k is an integer.

(1)

- (ii) Simplify $\frac{\sqrt{48} + 2\sqrt{27}}{\sqrt{12}}$, giving your answer as an integer.

(3)

- (b) Express $\frac{1 - 5\sqrt{5}}{3 + \sqrt{5}}$ in the form $m + n\sqrt{5}$, where m and n are integers.

(4)

(Total 8 marks)

Q16.

- (a) Show that $\frac{\sqrt{50} + \sqrt{18}}{\sqrt{8}}$ is an integer and find its value.

(3)

- (b) Express $\frac{2\sqrt{7} - 1}{2\sqrt{7} + 5}$ in the form $m + n\sqrt{7}$, where m and n are integers.

(4)

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(Total 7 marks)

Q17.

- (a) Express $(3 - \sqrt{5})^2$ in the form $m + n\sqrt{5}$, where m and n are integers.

(2)

- (b) Hence express $\frac{(3 - \sqrt{5})^2}{1 + \sqrt{5}}$ in the form $p + q\sqrt{5}$, where p and q are integers.

(4)

(Total 6 marks)

Q18.

- (a) Express $\frac{7 + \sqrt{5}}{3 + \sqrt{5}}$ in the form $m + n\sqrt{5}$, where m and n are integers.

(4)

- (b) Express $\sqrt{45} + \frac{20}{\sqrt{5}}$ in the form $k\sqrt{5}$, where k is an integer.

(3)

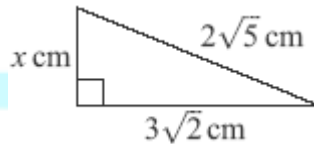
(Total 7 marks)

Q19.

- (a) Express $\frac{5 + \sqrt{7}}{3 - \sqrt{7}}$ in the form $m + n\sqrt{7}$, where m and n are integers.

(4)

- (b) The diagram shows a right-angled triangle.



The hypotenuse has length $2\sqrt{5}$ cm. The other two sides have lengths $3\sqrt{2}$ cm and x cm. Find the value of x .

(3)

(Total 7 marks)

Q20.

- (a) Express $5\sqrt{8} + \frac{6}{\sqrt{2}}$ in the form $n\sqrt{2}$, where n is an integer.

(3)

- (b) Express $\frac{\sqrt{2} + 2}{3\sqrt{2} - 4}$ in the form $c\sqrt{2} + d$, where c and d are integers.

(4)

(Total 7 marks)

Q21.

It is given that $x = \sqrt{3}$ and $y = \sqrt{12}$.

Find, in the simplest form, the value of:

- (a) xy ;

(1)

(b) $\frac{y}{x}$;

(2)

(c) $(x + y)^2$.

(3)

(Total 6 marks)

Q22.

(a) Express $\frac{\sqrt{5} + 3}{\sqrt{5} - 2}$ in the form $p\sqrt{5} + q$, where p and q are integers.

(4)

(b) (i) Express $\sqrt{45}$ in the form $n\sqrt{5}$, where n is an integer.

(1)

(ii) Solve the equation

$$x\sqrt{20} = 7\sqrt{5} - \sqrt{45}$$

giving your answer in its simplest form.

(3)

(Total 8 marks)

Q23.

(a) Express $\frac{\sqrt{63}}{3} + \frac{14}{\sqrt{7}}$ in the form $n\sqrt{7}$, where n is an integer.

(3)

(b) Express $\frac{\sqrt{7} + 1}{\sqrt{7} - 2}$ in the form $p\sqrt{7} + q$, where p and q are integers.

(4)

(Total 7 marks)

Q24.

(a) Simplify $(\sqrt{5} + 2)(\sqrt{5} - 2)$.

(2)

(b) Express $\sqrt{8} + \sqrt{18}$ in the form $n\sqrt{2}$, where n is an integer.

(2)

(Total 4 marks)

Q25.

- (a) Express $(4\sqrt{5} - 1)(\sqrt{5} + 3)$ in the form $p + q\sqrt{5}$, where p and q are integers.

(3)

- (b) Show that $\frac{\sqrt{75} - \sqrt{27}}{\sqrt{3}}$ is an integer and find its value.

(3)

(Total 6 marks)

Q26.

- (a) Simplify $(\sqrt{12} + 2)(\sqrt{12} - 2)$.

(2)

- (b) Express $\sqrt{12}$ in the form $m\sqrt{3}$, where m is an integer.

(1)

- (c) Express $\frac{\sqrt{12} + 2}{\sqrt{12} - 2}$ in the form $a + b\sqrt{3}$, where a and b are integers.

(4)

(Total 7 marks)

Q27.

The point A has coordinates $(6, 5)$ and the point B has coordinates $(2, -1)$.

- (a) Find the coordinates of the midpoint of AB .

(2)

- (b) Show that AB has length $k\sqrt{13}$, where k is an integer.

(3)

- (c) (i) Find the gradient of the line AB .

(2)

- (ii) Hence, or otherwise, show that the line AB has equation $3x - 2y = 8$.

(2)

- (d) The line AB intersects the line with equation $2x + y = 10$ at the point C . Find the coordinates of C .

(3)

(Total 12 marks)

Q28.

Express each of the following in the form $m + n\sqrt{3}$, where m and n are integers:

(a) $(\sqrt{3} + 1)^2$;

(2)

(b) $\frac{\sqrt{3} + 1}{\sqrt{3} - 1}$.

(3)

(Total 5 marks)

Q29.

(a) Using the binomial expansion, or otherwise, express $(1 + x)^4$ in ascending powers of x .

(3)

(b) (i) Hence show that $(1 + \sqrt{5})^4 = 56 + 24\sqrt{5}$.

(3)

(ii) Hence show that $\log_2(1 + \sqrt{5})^4 = k + \log_2(7 + 3\sqrt{5})$, where k is an integer.

(3)

(Total 9 marks)